

How OCA Code Gets Reviewed: Data & Methodology

A reproducible analysis of review independence across the Odoo Community Association's entire GitHub history.

Snapshot: **8 July 2026**. Author: Alexey Pelykh. All source data is public.

1. The question

For every code change merged into OCA, **who approved it** — an independent reviewer from a different company, a colleague at the author's own company, or nobody at all? And does that pattern change with the size of the company contributing?

This is a question about **review provenance**, measured from the public record. It is **not** a claim about the quality, intent, or content of any change. See §9 for what this analysis explicitly cannot say.

2. Data source

100% of the underlying data is public and comes from the GitHub REST/GraphQL API via the official `gh` CLI. No private data, no scraping of anything behind auth.

- **Scope:** all 261 repositories in the `OCA` GitHub organization.
- **Population:** every *merged* pull request in each repo, over the project's full lifetime.
- **Collected per PR:** number, author login, `createdAt`, `mergedAt`, `mergedBy`, and the full review list (each review's author, state, and `submittedAt`). For the subset with no cross-company approval, also the list of changed file paths.
- **Total collected:** 68,214 merged PRs across all years; **61,337** merged in 2017 or later (the analysis window).

Why 2017+

GitHub's Pull Request **Reviews API** did not exist before **September 2016**. PRs merged before then carry zero review objects *by construction* — not because they went unreviewed. Including them would fabricate a “100% unreviewed” prehistory that is a data artifact, not a finding. Every rate in this analysis is therefore computed over **PRs merged 2017 or later**, where review data is complete.

3. Attributing a contributor to a company

OCA's own charter states members participate "as individuals ... [so that] affiliations do not cloud the personal contributions." That is an ideal; this analysis measures the empirical reality, which requires linking a GitHub login to an employer. There is no canonical map (by design — see §6), so each login is resolved by **triangulation** of public signals, in priority order:

1. GitHub **profile** `company` **field** (primary; ~92% of resolved logins).
2. **Commit-author email domain** (e.g. `@tecnativa.com`).
3. **Username suffix** convention (e.g. `-tecnativa`, `-forgeflow`).
4. Public GitHub **organization membership**.

Company names are canonicalized (e.g. `Tecnativa S.L.` → `tecnativa`; `Eficent` → `forgeflow` after the known rebrand).

- **Coverage:** 79.3% of 2017+ merged PRs have a resolved author company (48,647 of 61,337).
- **Conservative rule — this is the load-bearing safeguard:** a login that cannot be resolved is treated as `unknown` and is **never** counted as sharing an author's company. An approval by an unknown reviewer counts as *independent/indeterminate*, never as self-serving. This means every "self-served review" figure is a **lower bound** — misattribution can only *understate* it.
- **Aggregate robustness:** because figures are org-level over thousands of PRs, a handful of individual misattributions cannot move a company's rate materially. A 55% self-serve rate over 10,148 PRs does not hinge on any one person's employer.

4. Definitions

For a merged PR authored by a contributor at company **X**, its review provenance is exactly one of:

Category	Definition
No review	Zero approving reviews (excluding the author, excluding bots).
Self-served	≥1 approving review, every approver resolved, all from company X , none external.
Independent	≥1 approving reviewer from a company other than X .
Indeterminate	≥1 approver whose company is <code>unknown</code> (cannot classify; excluded from rates).

- **Approving review** = a GitHub review whose *latest verdict-bearing* state by that reviewer is `APPROVED`. A reviewer who approves and later leaves a plain `COMMENT` still counts as approving (the comment does not retract the approval). `COMMENTED` / `PENDING` states never count as approvals.

- **No independent review** = *No review* + *Self-served* (the change reached `main` without any outside company signing off).
- **Bots** (`oca-git-bot`, CI accounts, etc.) are excluded from all review counts.
- **Self-of-reviewed** = $\text{Self-served} \div (\text{Self-served} + \text{Independent})$: of the PRs that got *any* review, what share was colleague-only.

Company-size gradient

Company size = count of distinct contributor logins resolved to that company. Orgs are bucketed (1 / 2–4 / 5–9 / 10–19 / 20+ people) and the self-serve rate reported per bucket. A one-person “company” cannot self-serve by definition (no colleague to approve).

Concentration

For authorship and for approving-reviews separately: each company’s share of the total, the top-1 and top-5 shares, the number of companies accounting for 50%, and the **Gini coefficient** (0 = perfectly even across all companies, 1 = one company does everything).

5. Module maintainership control (the fairness check)

A company reviewing its own PR is *expected and legitimate* when the PR touches a module that company **maintains** — OCA grants maintainers merge rights over their modules. To avoid mislabeling legitimate maintenance as capture, every no-independent-review PR was reclassified at **module** granularity:

- Each PR’s changed files → the module(s) it touched → each module’s **declared maintainers** (the `maintainers` key in the module’s `__manifest__.py`) → those maintainers’ companies.
- **(a) Excused**: the author’s company maintains every touched module (a colleague-maintainer counts — this is an *org-level* test, not a login test).
- **(b) Other-org turf**: the author’s company maintains *none* of the touched modules; they are another company’s declared territory. This is the only bucket that could indicate reviewing outside one’s remit.
- **(c) No declared maintainer**: the touched module declares no maintainer at all (~50% of cases; mostly older modules).

Result: of no-independent-review PRs, (a) excused ≈ 37%, (c) ambiguous ≈ 50%, **(b) other-org turf ≈ 14%** — with **no** company standing out as an outlier once maintainership is credited. The self-serve finding in §4 is a fact about *provenance*; this control shows most of it is legitimate territory maintenance, not cross-company imposition.

6. How an OCA merge actually works (sourced)

The reviewed-ness of a merge is a **social convention, not a mechanical gate**:

- `APPROVALS_REQUIRED` (default **2**) only sets the `approved` **label**; it is not referenced in the bot's merge task. — `OCA/oqa-github-bot/.../config.py`
- `/ocabot merge` checks exactly two things: **green CI** and that the triggering user has **push access OR is a declared maintainer of all modified addons. No approval count. No author-≠-merger check.** — `OCA/oqa-github-bot/.../tasks/merge_bot.py`
- CONTRIBUTING prose: “At least one of the review[s] above must be from a member of the PSC or having write access on the repository.” — a norm, not enforced. — `OCA/odoo-community.org/.../CONTRIBUTING.rst`
- No canonical login→company map is published, by charter design. — `OCA/odoo-community.org/.../Organization.rst`

A contributor with push access or self-declared maintainership can therefore merge their own PR with zero external approvals, and the tooling permits it.

7. Corrections applied (adversarial review)

The first-pass numbers were wrong in four ways, each caught and fixed before publication. They are listed here because a methodology that hides its own error history is not trustworthy:

1. **Pre-2017 removal** — see §2. Dropped a 6,900-PR artifact era.
2. **Approve-then-comment** — a reviewer who approved then commented was wrongly uncounted (~3.7% of PRs). Fixed to latest-verdict-bearing state.
3. **Login → org maintainership** — a PR merged by one employee into a module maintained by a *colleague* was first mis-scored as “other-org turf.” Corrected to an org-level test (§5), which moved ~1,140 PRs from (b) to (a) and roughly halved the apparent per-company capture.
4. **Current-snapshot maintainers** — see §8.

8. Merge-time verification

The maintainer list in a module's manifest is read at its *current* state, which could mask history: a company that merged another's module in 2019, then became its maintainer by 2026, would read as “excused” today. To test this, a stratified sample of **621** PRs (400 excused + 221 other-org-turf) was reclassified using each module's manifest **as it existed at that PR's actual merge commit**.

- **Hidden capture** (current-excused that was really other-org-turf at merge time): **1.5%** (95% CI ≤ ~3.2%).
- **False capture** (current-other-org-turf that was the company's own module at merge time): 4.1%.

The current-snapshot method is, if anything, slightly *generous* to the capture narrative — it does not hide it. The compression in §5 is real.

9. What this analysis CANNOT say

- **Nothing about the content or intent of any change.** “Self-served review” describes who approved, not whether the change was good, self-interested, or agenda-driven. Provenance ≠ motive.
- **Nothing about individuals.** All published figures are company-level aggregates. No per-person data is released.
- **21% of PRs are unattributed** (unknown company) and excluded from company rates.
- **Self-review is often legitimate** — see §5. High self-serve ≠ wrongdoing; it frequently reflects a company maintaining its own modules.
- **Correlation, not causation**, on the size gradient: large companies *can* self-review (they have colleagues); this does not establish that they do so to avoid scrutiny.

10. Reproduce it

Every figure derives from public GitHub data. To rebuild from scratch: 1. `gh pr list -R OCA/<repo> --state merged -L 200000 --json number,author,createdAt,mergedAt,mergedBy,reviews` for each of the 261 repos. 2. Resolve author/reviewer companies by the triangulation in §3. 3. Apply the definitions in §4 over PRs with `mergedAt year ≥ 2017`. The aggregate outputs are in the accompanying CSVs (§11).

11. Data dictionary

- `oca-review-by-org.csv` — per company (≥30 authored PRs, 109 companies): contributors, authored PRs, and the % breakdown across no-review / self-served / external, plus the module-controlled other-org-turf count.
 - `oca-review-by-year.csv` — per year: merged PRs and the % self-served / external / <2-approvals / zero-review. The erosion trend.
 - `oca-selfserve-by-orgsize.csv` — per company-size bucket: the self-serve rate. The size gradient.
 - `oca-concentration.csv` — authorship vs approving-reviews: top-1/top-5 share, companies-for-50%, Gini.
-

Appendix A — Concentration

dimension	total events	distinct orgs	top1 org pct	top5 orgs pct	orgs for 50pct	gini
authorship	48647	634	20.9	48.1	6	0.91
approving_reviews	96231	496	25.3	51.3	5	0.93

Appendix B — Review provenance by year

year	merged prs	pct lt2 approvals	pct zero review	pct self served	pct external reviewed
2017	3278	38.0	8.8	3.7	67.5
2018	3648	38.9	10.5	5.2	63.5
2019	4326	35.5	9.7	6.0	59.7
2020	5025	30.8	9.3	6.5	59.4
2021	6623	32.0	8.1	12.2	58.0
2022	7667	41.8	14.5	13.7	42.2
2023	8049	41.7	15.5	17.1	40.8
2024	8022	34.9	13.1	17.7	42.7
2025	9626	34.5	13.1	15.9	47.8
2026	5073	34.3	12.1	15.8	46.1

Appendix C — Self-serve rate by company size

org size	n orgs	authored prs	self served pct of authored	self served pct of reviewed
1	88	11708	0.1	0.1
2-4	40	7252	3.1	3.8
5-9	9	3188	13.0	15.1
10-19	5	11438	11.7	16.1
20+	3	13049	45.2	55.6

Appendix D — Review provenance by company (top 25 of 109)

org	contributors	authored prs	no review pct	self served pct	external reviewed pct	self of reviewed pct	b pct of authored
tecnativa	22	10148	13.0	55.4	27.6	66.7	5.8
forgeflow	18	4646	16.2	20.4	51.7	28.3	2.2
akretion	18	4006	19.8	6.5	70.6	8.4	0.8
camptocamp	28	2308	6.5	9.7	74.4	11.5	3.3
acsone	18	2306	31.6	5.3	58.6	8.3	1.9
dixmit	2	1221	15.4	2.5	73.4	3.3	4.3
moduon	6	1149	8.5	23.3	66.8	25.9	3.7
oca @camptocamp	1	970	27.7	0.0	67.3	0.0	1.9
escodoo	4	850	0.7	0.4	94.4	0.4	0.6
grap	2	756	17.6	4.8	74.7	6.0	0.8
ecosoft-odoo	2	674	6.7	6.8	70.8	8.8	4.7
open source integrators	7	622	24.0	8.7	60.9	12.5	7.2
vauxoo	21	593	55.6	8.8	32.5	21.2	1.0
takobi-online @innovyou	1	577	1.9	0.0	74.7	0.0	0.5
hunki enterprises	1	549	7.8	0.0	91.8	0.0	0.5
acsone s.a./ n.v.	1	547	20.8	0.0	77.9	0.0	3.1
gray matter logic	3	547	35.5	2.2	56.7	3.7	6.8
terrabit-ro / deltatech	1	496	95.4	0.0	4.6	0.0	15.5
opener b.v.	1	494	7.9	0.0	91.3	0.0	0.6
360erp	1	491	7.3	0.0	89.8	0.0	1.8
heliconia solutions pvt	1	477	1.7	0.0	96.0	0.0	1.3
agilebg	3	468	4.7	3.6	82.1	4.2	1.3
nexterp- romania	1	369	83.7	0.0	16.0	0.0	1.9
	1	367	6.0	0.0	93.7	0.0	0.5

org	contributors	authored prs	no review pct	self served pct	external reviewed pct	self of reviewed pct	b pct of authored
obertix, free software solutions							
trobz	7	345	2.6	0.0	95.4	0.0	1.2

Full 109-company table in [oca-review-by-org.csv](#). Percentages over each company's own authored PRs; 'self of reviewed' = colleague-only share of PRs that got any review.